Fault Ride-through for Cascaded Inverters Using Mixed Grid-forming and Grid-following Control

Rahul Mallik[1], Debjyoti Chatterjee[2], and Brian Johnson[2]
[1]Hanwha Q CELLS North America, Santa Clara, CA, USA
[2]Chandra Family Department of Electrical and Computer Engineering, The University of Texas at Austin

IEEE ENERGY CONVERSION CONGRESS & EXPO PHOENIX, ARIZONA, USA ✳ OCT. 20-24

U.S. DEPARTMENT OF ENERGY

The University of Texas at Austin
Chandra Department of Electrical and Computer Engineering
Cockrell School of Engineering

Modular converters are great..

- Offer higher reliability, standardized manufacturability
- Distributed voltage stress, devices with low breakdown V, low R_{dson}, higher efficiency
- Touch-safe PV inverters make easy serviceability for medium-voltage grid-side connection

Fault impacts decentralized controls

- During a fault, due to current limiting, there is a loss of synchronization between the cascaded modules and the grid
- This causes overcurrent both during fault onset and recovery

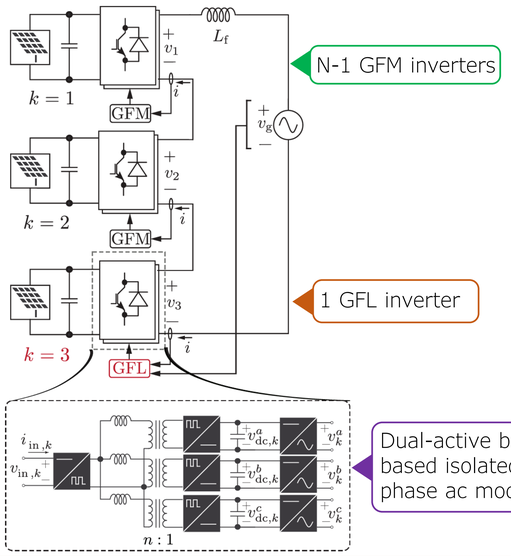
Leading to decentralized control

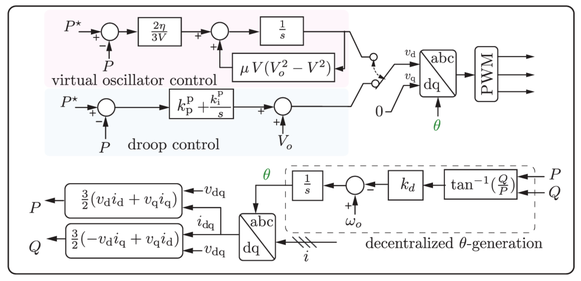
- Each module implements a "droop", resulting in a decentralized GFM operation
- no master controller and point-to-point wiring
- faster reaction to disturbances compared to centralized MMC-like controls

The solution

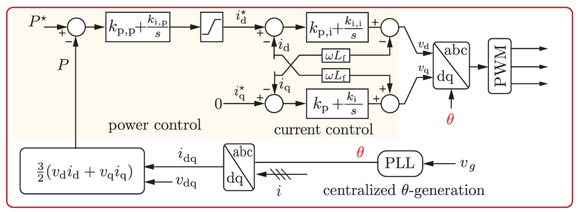
- Convert one series-connected module to be controlled as GFL inverter that regulates current during normal as well as fault operation

Proposed Solution: Mixed GFL-GFM sources in series connection





k=1, 2: Grid-forming Converter Control



k=3: Grid-following Converter Control

Controller Design for Mixed GFM-GFL system (small signal stability)

Timescale separation of the controllers yield closed form solutions for controller gains.

	Inv. Switching	Current Control	Power Control
GFL	ω_{sw}	$\omega^f_{cc} = \frac{R+k_p}{L}$	$\omega_{cc} = \frac{k_{ii}}{k_p+R}$	$\omega_p = \frac{k_{i,p}}{k_{p,p}+V_{nom}/1.5}$
	10 kHz	200 Hz	70 Hz	2.5 Hz

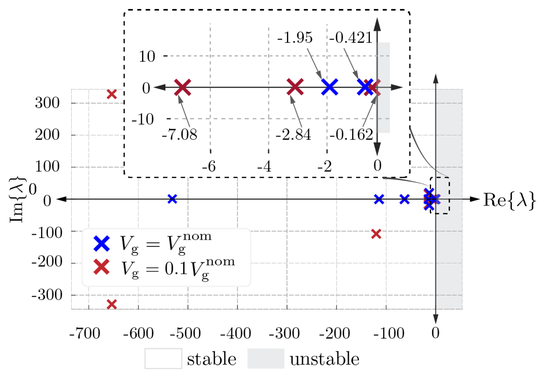
	Inv. Switching	Droop Control	Power Control
GFM	ω_{sw}	$\omega_\theta = k_d$	$\omega_p = \frac{k^i_p}{k^p_p}$
	10 kHz	10 Hz	3 Hz

GFL gains:

$$k_p = \epsilon^f_{cc}\omega_{sw}L - R,$$
$$k_i = \epsilon_{cc}\omega_{sw}(k_p + R),$$
$$k^p_i = \epsilon_p\omega_{sw}(k^p_p + (2/(3V_{nom}))).$$

GFM gains:

$$k^p_i = k^p_p\epsilon_p\omega_{sw}$$
$$k_d = \epsilon_\theta\omega_{sw}.$$

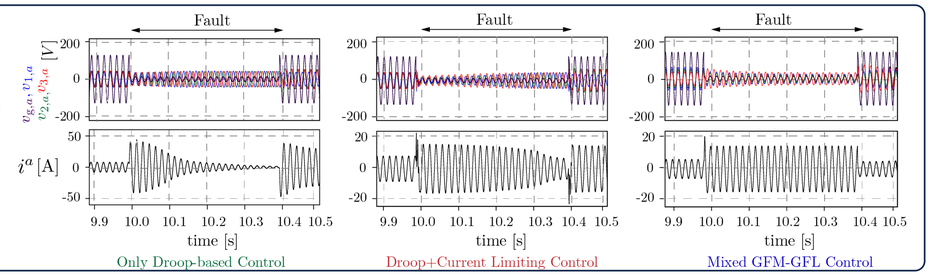


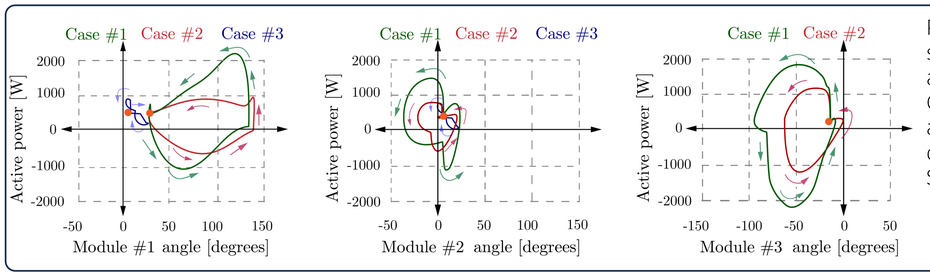
Using the design equations, small-signal stability is guaranteed for Nominal grid voltage, 10% undervoltage

Comparing Different FRT for Cascaded Converters

Decentralized controls →
- takes longer to sync after fault goes away
- large overcurrents at both fault onset and recovery.

Mixed GFM-GFL →
- tight regulation of current
- Quick synchronization



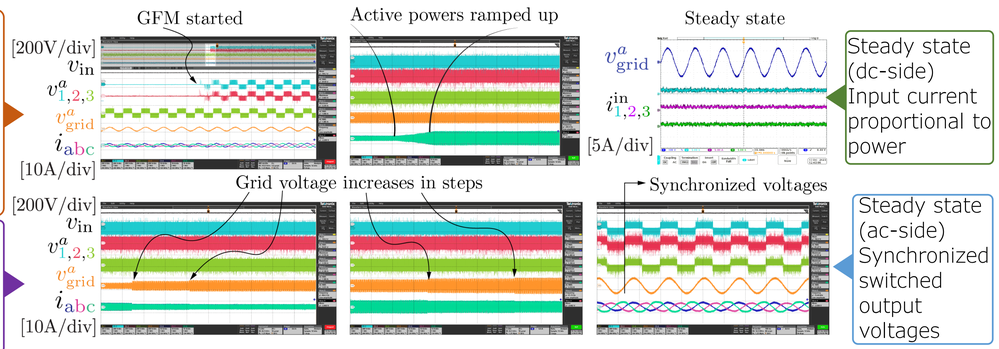


Pre- and post-fault steady states converge for each approach
Case #3 has the smallest area enclosed by the $P-\delta$ curve.
Smaller curve →
- smaller angle excursion
- lesser power fluctuation
- lesser overcurrents

Experimental Validation

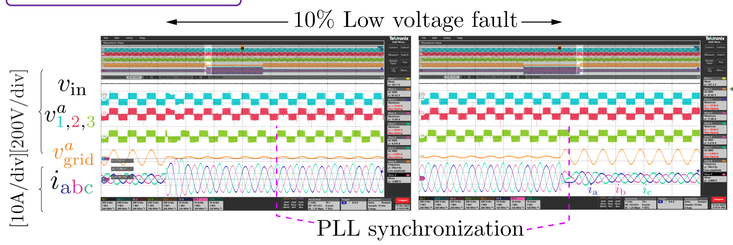
Start operation at low grid voltage , turn on the GFL unit, regulate the grid current, Turn on GFM units and they synchronize without any communication.

Ramp up the active powers
Increase the grid voltage to the rated value



Steady state (dc-side) Input current proportional to power

Steady state (ac-side) Synchronized switched output voltages



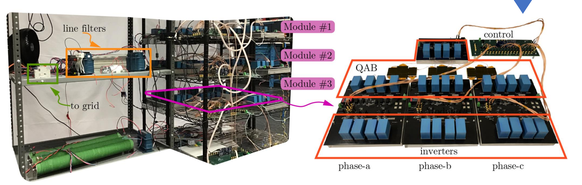
10% grid undervoltage ride-through with
- steady, controlled overcurrent during fault
- low overcurrent during and after fault
- rapid PLL synchronization

Experimental Setup shows three cascaded units,
- each module has three single phase inverters
- each inverter has an upstream isolated dc-dc converter

System specifications

	Symbol	Description	Value	Units
Ac system	S_{rated}	module VA rating	1500	VA
	N	modules in series	3	
	V_g	rms line-neutral grid voltage	90	V
	ω_{nom}	rated frequency	60	Hz
	L_f	inductance	2.4	mH
	R_f	resistance	4.2	Ω

	Symbol	Description	Value	Units
QAB dc-links	V^n_{dc}	nominal dc voltage	100	V
	ω_{sw}	switching frequency	100	kHz
	L	inductance	26	μH
	n	turns ratio	0.5	
	C_{pv}	input capacitance	660	μF
	C_{dc}	dc-link capacitance	200	μF
	$k_{p,dc}$	proportional gain	2.05	Ω^{-1}
	$k_{i,dc}$	integral gain	1290	$(\Omega s)^{-1}$



References

X. Hou, K. Sun, X. Zhang, Y. Sun and J. Lu, "A Hybrid Voltage/Current Control Scheme With Low-Communication Burden for Grid-Connected Series-Type Inverters in Decentralized Manner," in IEEE TPEL, Jan 2022

Acknowledgement

This material is based upon work supported by the U.S. DOE's Office of Energy Efficiency and Renewable Energy (EERE) under Solar Energy Technologies Office (SETO) Agreement Number EE0008346 and Award Number 38637